

VB language: operators, if, Select Case, loops, arrays, functions**★ Golden**

One college period · 40–45 minutes · Programming using Visual Basic

What you will learn

- Write If...Then and Select Case
- Write For...Next, Do While, Do Until
- Declare arrays and small functions
- Translate formulae into VB

```
If marks ≥ 40 Then
    lblR.Caption = "Pass"
Else
    lblR.Caption = "Fail"
End If
```

```
Select Case ch
    Case 1: lbl.Caption = "Add"
    Case 2: lbl.Caption = "Sub"
    Case Else: lbl.Caption = "Invalid"
End Select
```

```
Dim i As Integer, s As Integer
s = 0
For i = 0 To 50 Step 2
    s = s + i      ' even numbers 0-50
Next i
```

```
' sum and average of 10 inputted numbers
Dim i As Integer, n As Double, t As Double
t = 0
For i = 1 To 10
    n = Val(InputBox("Number " & i))
    t = t + n
Next i
MsgBox "Sum=" & t & " Avg=" & t / 10
```

Maths	VB
$x = 3ab^2 / c$	<code>x = 3 * a * b ^ 2 / c</code>
$Y = (4/3)\pi r^3$	<code>Y = (4 / 3) * 3.14 * r ^ 3</code>
$Y = (a+b)^n$	<code>Y = (a + b) ^ n</code>
$a = \frac{1}{2} \text{ base} \cdot \text{height}$	<code>a = 0.5 * base * height</code>

Built-in functions: Val, Str, Left, Right, Mid, Len, UCase, LCase, Int, Rnd, Date, Sqr, Abs, Format. User-defined: Function Area(r As Double) As Double : Area = 3.14 * r * r : End Function. Errors: syntax, run-time, logical. On Error GoTo handler.

HOW BIEK ASKS THIS

2026 programs: even 0–50; sum and average of ten numbers. Long: arrays; loops; built-in function with a program.

اردو میں سمجھیں

VB میں If...Then، Select Case، For...Next اور Do While لوپ ہیں۔ ^ طاقت ہے۔ Val ٹیکسٹ کو عدد بناتا ہے۔

CHECK YOURSELF

1. Even numbers 0–50: Step?

2. Val("12a") in VB6?

Answers: 1. 2, start 0 · 2. 12 (stops at first non-numeric, classic VB)

MCQs

- Comparison operators are also called:
 - Comparators
 - Comparables
 - Relations
 - Relational
- The last statement in a Function is often:
 - Footer
 - Header
 - End Function / Return value assignment
 - Call
- A function returns a:
 - form
 - value
 - only a MsgBox
 - table

Key: 1=d, 2=c, 3=b

Short questions (3 marks each)

- Write the structure of Select Case.
- Write a VB program to generate even numbers 0–50.
- Write a VB program to calculate sum and average of ten numbers.
- Convert $Y = (4/3)\pi r^3$ to VB.
- What is an error? Define its types.

Model short answers

- Select Case expr : Case v1: ... : Case Else: ... : End Select
- For i = 0 To 50 Step 2 : Print i : Next
- Loop 10 InputBox, add to total, divide by 10.
- $Y = (4 / 3) * 3.14 * r ^ 3$
- Mistake in a program. Syntax (grammar), run-time (divide by zero), logical (wrong formula, runs but answer wrong).

Long questions (board Section C)

1. Explain arrays in VB with types and an example. OR Explain loops with two types and programs. OR Built-in functions with a program.